

Mathematics A

Unit 1 • Chapter OL-T49 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 26 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T49 • Expertise Q16+ • 5 items • 26 marks

Chapter	Items	Marks	Starts
01. OL-T49 Sine, Cosine Rule & Area of Triangles	5	26	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Sine, Cosine Rule & Area of Triangles

Unit 1 • Expertise • 26 marks • 5 item(s)

June 2025 | 4MA1-1H Q20 | 5 marks

Use two non-right triangles sharing a diagonal

June 2023 | 4MA1-2H Q22 | 5 marks

Combine non-right trigonometry with circle or elevation geometry

June 2022 | 4MA1-1H Q18 | 5 marks

Find an acute angle using the sine rule

January 2022 | 4MA1-2H Q23 | 6 marks

Resolve the ambiguous sine-rule angle

January 2023 | 1H Q16 | 5 marks

Find a side using the sine rule

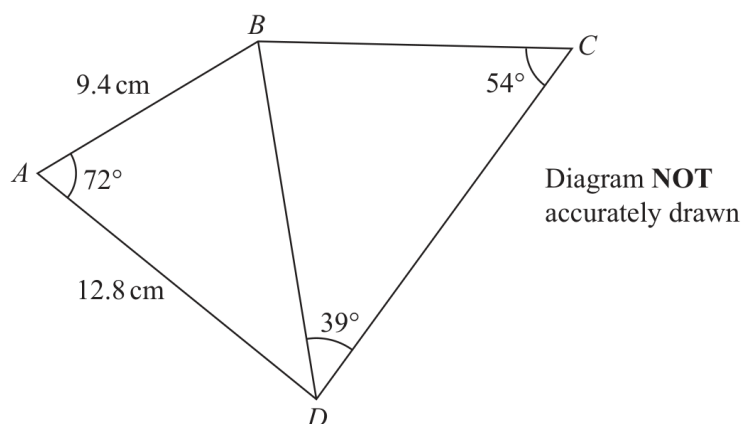
Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q20

5 marks

20



Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

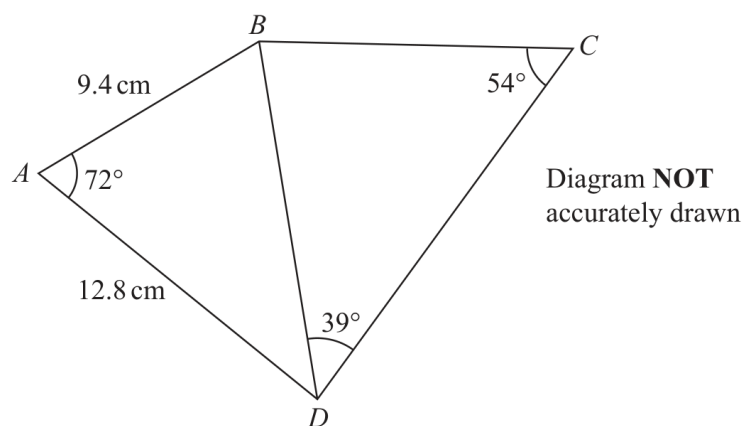
(Total for Question 20 is 5 marks)

WORKING SPACE

4MA1-1H Q20

5 marks

20



Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

Cosine rule followed by sine rule

5 marks

Find the shared side

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8) \cos 72^\circ, \quad BD = 13.335597 \dots$$

Use the sine rule

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ} \Rightarrow BC = \frac{13.335597 \dots \sin 39^\circ}{\sin 54^\circ} = 10.3735 \dots$$

Final answer

$$BC = 10.4 \text{ cm (3 s.f.)}$$

Dependency check Retain the unrounded value of BD for the second triangle.

4MA1-2H Q22

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

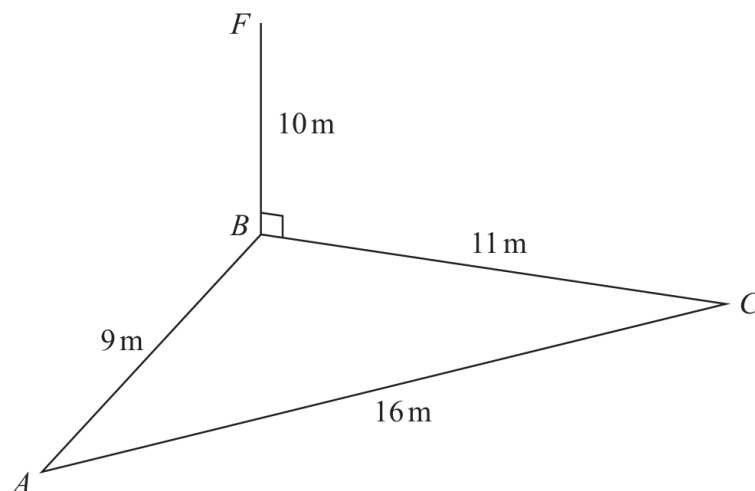


Diagram **NOT** accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

○

Continue your method

5 marks

WORKING SPACE

4MA1-2H Q22

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

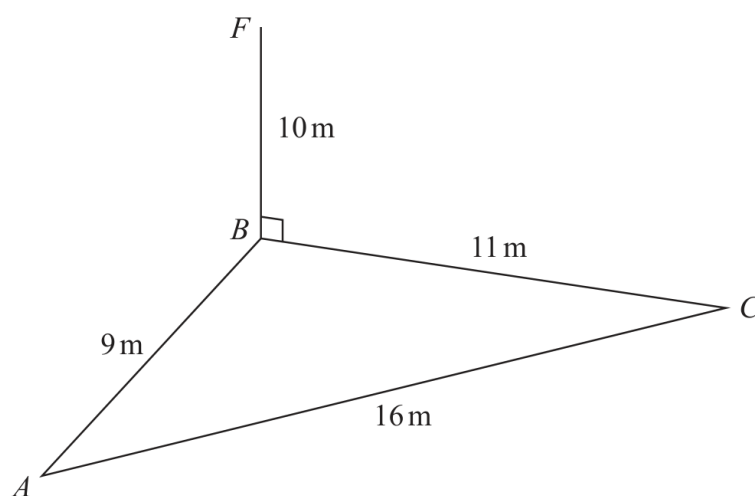


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Work out the size of the angle of elevation of the point F from the point D

Give your answer correct to one decimal place.

○

Angle of elevation from a perpendicular foot

5 marks

Find the ground angle

$$\cos \angle BCA = \frac{11^2 + 16^2 - 9^2}{2(11)(16)} = \frac{37}{44} \Rightarrow \angle BCA \approx 32.763^\circ.$$

Find BD

$$BD = 11 \sin 32.763^\circ \approx 5.95 \text{ m.}$$

Use the vertical flagpole

$$\tan \angle FDB = \frac{BF}{BD} = \frac{10}{5.95}.$$

Evaluate

$$\angle FDB \approx 59.2^\circ.$$

Final answer

$$59.2^\circ$$

4MA1-1H Q18 Q18

5 marks

18 Here is triangle ABC

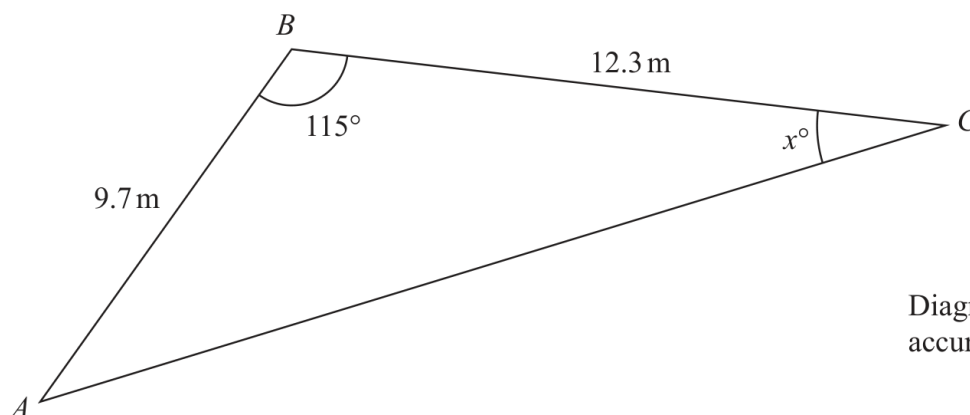


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q18 Q18

5 marks

18 Here is triangle ABC

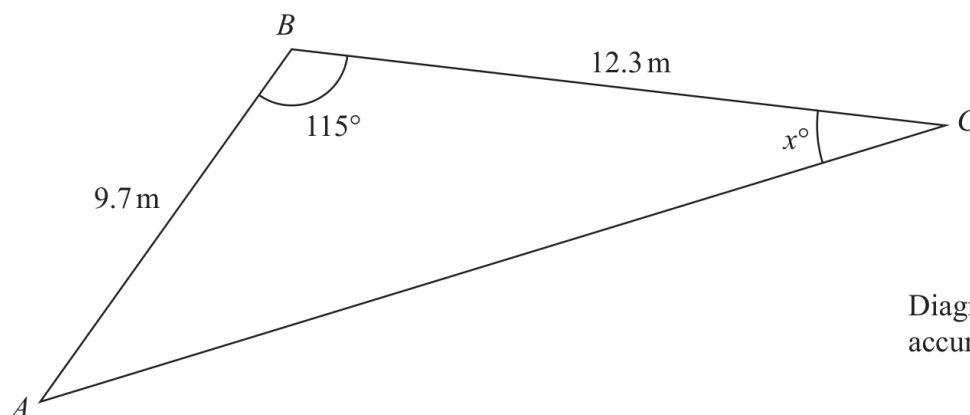


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$

Chapter: Sine, Cosine Rule & Area of Triangles

5 marks

Q18 – Chapter: Sine, Cosine Rule & Area of Triangles

Family: Use the sine rule • Variant: Find an acute angle using the sine rule

MethodBy the cosine rule, $AC^2 = 9.7^2 + 12.3^2 - 2(9.7)(12.3)\cos(115 \text{ degrees})$.*Why: The two known sides enclose the known 115-degree angle.***Method** $AC^2 = 346.2... , \text{ so } AC = 18.606... \text{ m.}$ *Why: Evaluate the cosine-rule expression without premature rounding.***Method**Using the sine rule, $\sin(x)/9.7 = \sin(115 \text{ degrees})/18.606... .$ *Why: Side AB = 9.7 is opposite angle x, and AC is opposite 115 degrees.***Method**Therefore $\sin(x) = 9.7 \sin(115 \text{ degrees})/18.606... = 0.472... .$ *Why: Rearrange the sine-rule equation for $\sin(x)$.***Method** $x = \sin^{-1}(0.472...) = 28.2 \text{ degrees to 3 significant figures.}$ *Why: Take the inverse sine in degree mode and round once at the end.***Final answer****angle:** $x = 28.2 \text{ degrees}$

4MA1-2H Q23

6 marks

23 The diagram shows triangle PQR

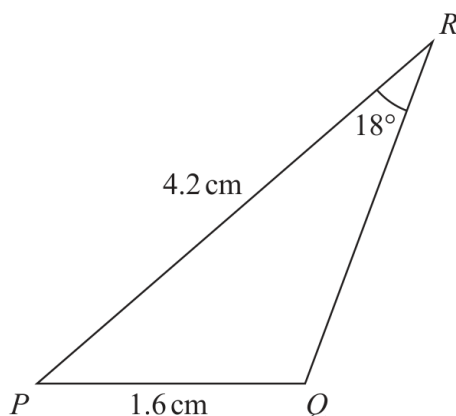


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

Continue your method**6 marks**

WORKING SPACE

4MA1-2H Q23

6 marks

23 The diagram shows triangle PQR

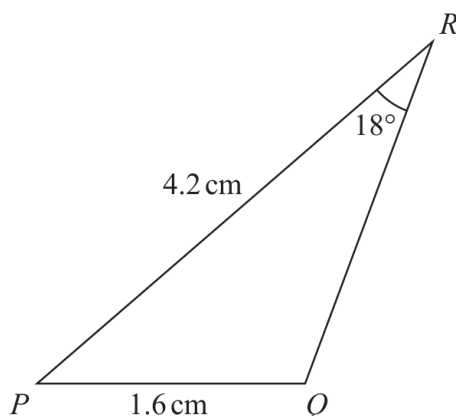


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$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

Chapter: Sine, Cosine Rule & Area of Triangles

6 marks

Q23 – Chapter: Sine, Cosine Rule & Area of Triangles

Family: Use the sine rule • Variant: Resolve the ambiguous sine-rule angle

MethodUsing the sine rule, $\sin Q/4.2 = \sin 18^\circ/1.6$, giving $Q = 54.2^\circ$ or 125.8° .*Why: The sine equation has an acute and an obtuse possibility.***Method**The question says Q is obtuse, so $Q = 125.8^\circ$.*Why: Select the permitted branch.***Method** $P = 180 - 125.8 - 18 = 36.2^\circ$.*Why: Angles in a triangle total 180° .***Method**

The corresponding side RQ is 3.0585... cm.

*Why: Use the cosine rule or the sine rule after finding P.***Method** $\text{Area} = \frac{1}{2} \times 4.2 \times 1.6 \times \sin 36.2^\circ$.*Why: Use two sides and the included angle.***Method** $\text{Area} \approx 1.98 \text{ cm}^2$ (3 s.f.).*Why: Round the calculator value.***Final answer****area:** 1.98 cm^2

- 16 Here is a shape formed from two triangles ABC and CDE
 ACD and BCE are straight lines.

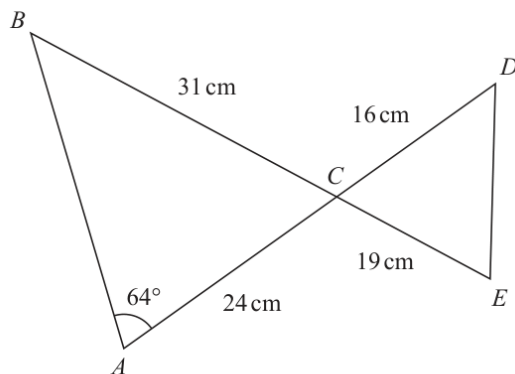


Diagram **NOT**
 accurately drawn

$$AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$$

$$\text{Angle } BAC = 64^\circ$$

Work out the length of DE

Give your answer correct to 3 significant figures.



January 2023 | 1H | Question 16 | 5 marks

Chapter OL-T49: Sine, Cosine Rule & Area of Triangles

..... cm

(Total for Question 16 is 5 marks)

Answer reference | January 2023 | Question 16

Family: Use the sine rule • Variant: Find a side using the sine rule

16	$\frac{\sin ABC}{24} = \frac{\sin 64}{31}$ oe		5	M1
	$(ABC =) \sin^{-1}\left(\frac{24 \times \sin 64}{31}\right) (= 44. \dots)$			M1
	$180 - "44. \dots" - 64 (= 71.9 \dots)$			M1 accept 72
	$(DE^2 =) 16^2 + 19^2 - 2 \times 16 \times 19 \times \cos "71.9 \dots"$ or $(DE =) \sqrt{16^2 + 19^2 - 2 \times 16 \times 19 \times \cos "71.9 \dots"}$ or $(DE =) \sqrt{617 - 181.8 \dots}$ or $\sqrt{428.166 \dots}$			M1 for DE^2 or DE
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	20.7		A1 awrt 20.7
				Total 5 marks

Worked solution

January 2023 | Question 16 | 5 marks

Family: Use the sine rule • Variant: Find a side using the sine rule

Official final mark-scheme pages are embedded immediately after this question.